EDIDIONG EKONG

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SUMMARY

Computer Engineering student at UT San Antonio with hands-on experience across embedded systems, digital and VLSI design, and full-stack software. Builds and debugs everything from C/C++ on microcontrollers and Verilog RTL to cloud-deployed Python services, and leads two IEEE student chapters.

EDUCATION

The University of Texas at San Antonio (UT San Antonio) | San Antonio, TX — Cumulative GPA: 3.98
Bachelor of Science in Computer Engineering — Expected Graduation: Spring 2027
Relevant Coursework: Embedded Systems Design, VLSI Design, Senior Design, Systems Programming for Engineers, Computer Organization & Architecture, Digital Systems Design, C++ & Data Structures, Circuits I

TECHNICAL SKILLS

- Languages: C, C++, Verilog, Python, JavaScript
- Embedded: Embedded C, Zephyr RTOS, Bluetooth Low Energy, TinyML
- EDA & CAD: Vivado, Cadence Virtuoso, Cadence Innovus, Altium Designer, KiCAD, Fusion360
- Microcontrollers: MSP430, STM32 (Wb55, Nucleo), ESP32, PIC16F1829
- Developer Tools & Platforms: Git, Code Composer Studio, discord.py, Supabase, Railway

EXPERIENCE

Spequlo | Software Engineer — May 2025 – Present

- Built Dipersa, an internal Discord bot that manages ClickUp tasks via natural-language requests and slash commands, using Python, LLM intent classification, discord.py, and Supabase
- Refactored the bot's classification pipeline to support multi-intent messages, reply-chain context, and structured JSON output, backed by typed error handling
- Developing a multi-tenant TikTok posting API (connected accounts, API key issuance) and an automated content pipeline with a control dashboard, deployed on Railway

PROJECTS

Lower-Limb Exoskeleton Control | Senior Design, ISABEL UT San Antonio — August 2026 – Present

- Designing the operator interface for a Lowerlimb Exoskeleton controller built on the OpenExo platform by ISABEL that displays real-time power, sensor, and joint/actuator diagnostic data
- Implementing wireless telemetry on an STM32WB55 running Zephyr RTOS over Bluetooth Low Energy

Real-Time Spoken-Digit Recognition | C, MSP430FR5994 — Fall 2026

- Implementing a neural network in C on an MSP430FR5994 that classifies spoken digits captured from a microphone module in real time

4x4 Systolic Array | Verilog, Cadence Virtuoso, Cadence Innovus — Fall 2026

- Designing a 4x4 systolic array for matrix multiplication in Verilog and carrying it through layout and place-and-route in Cadence Virtuoso and Innovus

Hand-Crank AC Generator – Fusion 360 — Spring 2026 - Fall 2026

- Led a team that designed and 3D-printed an AC generator producing 20 W of AC power for the IEEE PES at UT San Antonio Model Substation project

LEADERSHIP & ACTIVITIES

IEEE Power Electronics Society (PELS) at UT San Antonio| President — June 2026 – Present

- Leading the chapter's petition for official IEEE Student Branch Chapter status
- Run hands-on technical workshops, including intro to power, soldering, and intro to microcontrollers, for members and UT San Antonio students.
- Created a Junior Officer program in which new members complete month-long hardware projects and run events and workshops

IEEE Power and Energy Society (PES) at UT San Antonio | Vice Chair — January 2026 – Present

- Mentor junior officers and oversee the Junior Officer Program ran by the PES Officer Developer
- Maintain the chapter's IEEE Student Branch Chapter standing and its relationship with the local IEEE section